

Table 4: Results (Hit@1) on cold/warm item scenario. A-LLMRec (SBERT) is a variant of A-LLMRec that uses q instead of e for inference.

	Movies and TV		Video Games		Beauty	
	Cold	Warm	Cold	Warm	Cold	Warm
SASRec	0.2589	0.6787	0.1991	0.5764	0.1190	0.6312
MoRec	0.2745	0.4395	0.2318	0.4977	0.2145	0.5425
CTRL	0.1517	0.3840	0.2074	0.2513	0.1855	0.4711
RECFORMER	0.3796	0.5449	0.3039	0.5377	0.3387	0.5133
TALLRec	0.2654	0.2987	0.3950	0.4897	0.5462	0.6124
A-LLMRec	0.5714	0.6880	0.4263	0.5970	0.5605	0.6414
A-LLMRec (SBERT)	0.5772	0.6802	0.4359	0.5792	0.5591	0.6405

from user-item interactions (i.e., LLM-Only and TALLRec), implying that the collaborative knowledge is crucial for improving the performance of recommendation in general. 2) We observe that MLP-LLM, which replaces the alignment module of A-LLMRec with a simple MLP, underperforms A-LLMRec. This implies that bridging between CF-RecSys and LLM is a challenging problem and that our proposed two-stage alignment module is beneficial. 3) ‘LLM-Only’ performs the worst among the LLM-based models, implying that naively adopting an LLM based on a prompt designed for the recommendation task is not sufficient. Note that the prompt used by ‘LLM-Only’ is exactly the same as the prompt shown in Figure 3 without user representation and item embeddings. This again demonstrates the importance of incorporating collaborative knowledge into the LLM for improving the recommendation performance. 4) While TALLRec fine-tunes the LLM for the recommendation task, it underperforms a collaborative filtering model, SASRec. This highlights that the text information alone may not generate sufficient knowledge for capturing collaborative knowledge effectively even with fine-tuning the LLM. This again demonstrates the superiority of our alignment module. 5) Although the modality-aware models (MoRec and CTRL) use SASRec as the backbone CF-RecSys, they underperform SASRec. Moreover, RECFORMER struggles to outperform SASRec despite using Longformer for item text attributes, due to the emphasis on textual information in similarity matching between user and item sentences. This shows that the modality knowledge might hinder the learning of collaborative knowledge, leading to performance degradation.

5.2.2 Cold/Warm Item Scenarios. This section evaluates the models under cold/warm item scenarios. Items are labeled as ‘warm’ if they belong to the top 35% of interactions, while those in the bottom 35% are labeled as ‘cold’ items. After training each model using all the available data in the training set, we separately evaluate cold and warm items in the test set (Table 4). We make the following observations: 1) A-LLMRec outperforms all other baselines across both scenarios, which demonstrates that our alignment network indeed allows the LLM to understand and utilize the collaborative knowledge. 2) On the other hand, TALLRec outperforms SASRec only under cold scenario, whereas SASRec outperforms TALLRec only under warm scenario. This demonstrates the importance of capturing both the collaborative knowledge and the text information to excel in both cold/warm scenarios. 3) A-LLMRec (SBERT) outperforms A-LLMRec under the cold item scenario, while A-LLMRec generally outperforms A-LLMRec (SBERT) under the warm item scenario. As discussed in Section 4.1.4, this implies that the

Table 5: Results (Hit@1) on cold user scenario.

	Movies and TV	Video Games	Beauty
SASRec	0.2589	0.4048	0.4459
MoRec	0.3918	0.3572	0.4815
CTRL	0.2273	0.1737	0.3902
RECFORMER	0.4481	0.3989	0.4644
TALLRec	0.2143	0.3895	0.5202
MLP-LLM	0.4909	0.3960	0.5276
A-LLMRec	0.5272	0.4160	0.5337

Table 6: Results (Hit@1) on the few-shot training scenario on various datasets (K : num. users in the training set).

	K	SASRec	MoRec	TALLRec	A-LLMRec	A-LLMRec (SBERT)
Movies and TV	256	0.2111	0.2208	0.1846	0.2880	0.2963
	128	0.1537	0.1677	0.1654	0.2518	0.2722
Video Games	256	0.1396	0.1420	0.2321	0.2495	0.2607
	128	0.1089	0.1157	0.1154	0.1608	0.1839
Beauty	256	0.2243	0.2937	0.3127	0.3467	0.3605
	128	0.1813	0.2554	0.2762	0.3099	0.3486

joint collaborative-text embedding obtained from the text encoder given the text information (i.e., $q_i = f_T^{enc}(Q_i)$) is more useful than that obtained from the item encoder given the item embedding (i.e., $e_i = f_I^{enc}(E_i)$).

5.2.3 Cold User Scenarios. Besides evaluations under the cold item scenario, we additionally conduct evaluations under the cold user scenario (Table 5). To simulate the cold user scenario, we sample users who have interacted with exactly three items, where the last item in the sequence serves as the test set. Then, we use the models trained on the entire set of users except for the sampled users to perform inference on the sampled users. We observe that A-LLMRec consistently outperforms other models in the cold user scenario, while SASRec struggles to perform well, especially on a large dataset, i.e., Movies and TV, due to the lack of collaborative knowledge from users. Moreover, LLM-based models demonstrate superior performance in handling cold users as text information becomes useful under cold scenarios.

5.2.4 Few-shot Training Scenario. To investigate the impact of unseen/new items on recommendation models, we conduct experiments on a few-shot training scenario where the number of users in the training set is extremely limited to only K users, i.e., K -shot (Table 6). Under this scenario, we expect the models to encounter a large amount of unseen/new items at the inference stage, which would make it hard to provide accurate recommendations. We have the following observations: 1) A-LLMRec outperforms all other baselines under the few-shot scenario. Despite being trained with extremely small amount of users, A-LLMRec relies on CF-RecSys to capture the collaborative knowledge, which is combined with the textual knowledge of items, leading to superior performance in few-shot learning. 2) A-LLMRec (SBERT) outperforms A-LLMRec, implying again that using the text encoder to extract the joint text-collaborative knowledge is useful when items lack interactions. 3) Under the few-shot scenario, LLM-based models outperform the CF-RecSys, i.e., SASRec, due to the textual understanding of LLM, which helps extract information from the text of the unseen item, while CF-RecSys suffers from the lack of collaborative knowledge regarding unseen/new items.

Table 7: Results (Hit@1) on a cross-domain scenario (i.e., Pre-trained: Movies and TV, Evaluation: Video Games).

	SASRec	MoRec	RECFORMER	TALLRec	A-LLMRec	A-LLMRec (SBERT)
Movies and TV → Video Games	0.0506	0.0624	0.0847	0.0785	0.0901	0.1203

5.2.5 Cross-domain Scenario. To further investigate the generalization ability of A-LLMRec, we evaluate the models on the cross-domain scenario, where the models are evaluated on datasets that have not been used for training (Table 7). Specifically, we pre-train the models on the Movies and TV dataset and perform evaluations on the Video Games dataset. We have the following observations: **1)** A-LLMRec outperforms all the baselines in the cross-domain scenario, and A-LLMRec (SBERT) particularly performs well. This is again attributed to the text encoder that becomes useful when collaborative information is lacking. **2)** SASRec underperforms modality-aware models and LLM-based models, indicating that using textual knowledge is crucial for the cross-domain scenario due to the lack of collaborative information.

5.3 Ablation Studies

In this section, we show ablation studies for our model. We mainly analyze the effect of each component in A-LLMRec regarding Stage-1 (Section 5.3.1) and Stage-2 (Section 5.3.2).

Table 8: Ablation studies on Stage-1 of A-LLMRec (Hit@1).

Ablation	Movies and TV	Beauty	Toys
A-LLMRec	0.6237	0.5809	0.3336
w/o $\mathcal{L}_{\text{matching}}$	0.5838	0.5548	0.3225
w/o $\mathcal{L}_{\text{item-recon}} \& \mathcal{L}_{\text{text-recon}}$	0.5482	0.5327	0.3204
w/o $\mathcal{L}_{\text{rec}}$	0.6130	0.5523	0.1541
Freeze SBERT	0.6173	0.5565	0.1720

5.3.1 Effect of Components in Stage-1. This section presents the experimental results showing the benefit of each component during the Stage-1. Across all datasets, the exclusion of any loss resulted in decreased performance. We make the following observations: **1)** Removing $\mathcal{L}_{\text{matching}}$ from in Equation 2 results in a significant performance decline across all datasets. This implies that the alignment between the item and the text information is effective and that the LLM can comprehend item textual information in joint collaborative-text embeddings to enhance recommendation capabilities. **2)** Removing $\mathcal{L}_{\text{item-recon}}$ and $\mathcal{L}_{\text{text-recon}}$ leads to performance drop, owing to the risk of over-smoothed representations (i.e., $\mathbf{e} \approx \mathbf{q}$), as discussed in Section 4.1.1. **3)** We observe that removing $\mathcal{L}_{\text{rec}}$ leads to performance drop. Since $\mathcal{L}_{\text{rec}}$ is introduced to explicitly incorporate the collaborative knowledge while informing the model about the recommendation task, the performance drop indicates the reduction of collaborative knowledge between items and users, which is crucial for recommendation tasks. **4)** Lastly, we kept SBERT frozen while training A-LLMRec. We observe that freezing SBERT leads to poor performance across all datasets. This implies that fine-tuning SBERT facilitates the text embeddings to adapt to the recommendation task.

5.3.2 Effect of the Alignment method in Stage-2. Recall that a user representation and item embeddings are injected to the LLM prompt

Table 9: Ablation study on Stage-2 of A-LLMRec (Hit@1).

Row	Ablation	Movies and TV	Video Games	Beauty	Toys
(1)	A-LLMRec	0.6237	0.5282	0.5809	0.3336
(2)	A-LLMRec w/o user representation	0.5925	0.5121	0.5547	0.3217
(3)	A-LLMRec w/o joint embedding	0.1224	0.4773	0.5213	0.2831
(4)	A-LLMRec with random joint embedding	0.1200	0.4729	0.5427	0.0776

as shown in Figure 3. In this section, we verify the benefit of injecting them into the prompt (rows (2-4) in Table 9). We have the following observations: Across all datasets, **1)** the absences of either the user representation (row (2)) or the joint embedding (row (3)) from the prompt led to a reduction in performance. Notably, the exclusion of the joint embedding results in a more substantial decrease, underscoring its significant role in transferring collaborative knowledge. Moreover, as joint embeddings also capture the textual information about items, their exclusion is particularly detrimental. **2)** When we replace the joint embedding with a randomly initialized embedding (row (4)), which means A-LLMRec is trained with item embeddings without collaborative knowledge, we observe performance degradation across all datasets. This indicates the importance of leveraging the collaborative knowledge for recommendation.

5.4 Model Analysis

5.4.1 Train/Inference Speed. Recall that A-LLMRec requires the fine-tuning of neither the CF-RecSys nor the LLM. Specifically, A-LLMRec efficient in that the alignment network is the only trainable neural network, while TALLRec [2] requires the fine-tuning of the LLM with LoRA. In this section, we compare the training and the inference time of A-LLMRec and TALLRec. As for the training time, we measured the total time spent until the end of training, and as for the inference time, we measured the time spent per mini-batch. Table 10 shows that A-LLMRec exhibits significantly faster training and inference time compared with TALLRec. Notably, a more substantial improvement is observed in training time, since A-LLMRec does not require the LLM to be fine-tuned unlike TALLRec, which demonstrates the applicability of LLM in large-scale recommendation datasets. Moreover, the faster inference time demonstrates the practicality of A-LLMRec in real-world scenarios, especially in the context of real-time recommendation services where inference time is critically important.

5.4.2 Training with all items in each sequence. Recall that for efficiency in training, we used only the last item of each user sequence when optimizing the final loss in Stage-1 (Equation 6) and Stage-2 (Equation 8) of A-LLMRec. In this section, we report the recommendation performance in terms of Hit@1 and train/inference speed when using all items in each user sequence for optimization (see A-LLMRec_{all} in Table 10). We observe that as expected the recommendation performance is further improved when using all items in each user sequence. However, considering that the training time also increased approximately 3 times, the improvement seems marginal. It is important to note that since vanilla A-LLMRec is trained based on only the last item in each user sequence, there is a large amount of unseen/new items that appear in the test set⁴. However,

⁴About 13% of items are unseen during training in the Beauty dataset.

Table 10: Train/Inference time comparison (Beauty dataset).

	Train time (min)	Inference time (sec/batch)	Hit@1
TALLRec	588.58	3.36	0.5542
A-LLMRec	232.5	1.98	0.5809
A-LLMRec _{all}	643.33	1.98	0.6002

Table 11: Results showing A-LLMRec is model-agnostic.

Model	Beauty	Toys
SASRec	0.5298	0.2359
A-LLMRec (SASRec)	0.5809	0.3336
NextIttNet	0.4231	0.1415
A-LLMRec (NextIttNet)	0.5642	0.3203
GRU4Rec	0.4131	0.1673
A-LLMRec (GRU4Rec)	0.5542	0.3089
NCF	0.2957	0.1849
A-LLMRec (NCF)	0.5431	0.3263

vanilla A-LLMRec still showed comparable performance with A-LLMRec_{all}, implying the generalization ability of A-LLMRec.

5.4.3 A-LLMRec is Model-Agnostic. Although A-LLMRec adopts SASRec as the backbone CF-RecSys, it can be replaced with any existing collaborative filtering recommender systems, thanks to the model-agnostic property. Hence, we adopt three other collaborative filtering recommender systems including two sequential recommenders (i.e., NextIttNet and GRU4Rec), and one non-sequential recommender (i.e., NCF) to A-LLMRec. We make the following observations from Table 11. 1) Adopting the SASRec backbone performs the best, which is expected since SASRec outperforms other CF-RecSys in their vanilla versions. This implies that transferring high-quality collaborative knowledge can enhance the performance of A-LLMRec. 2) Adopting A-LLMRec to any backbone improves the performance of the vanilla model. This implies that if the SOTA model changes in the future, our framework has the potential to further improve performance by replacing the existing CF-RecSys in the framework. 3) We observe that while the performance difference between SASRec and NCF is nearly double when they operate as standalone CF-RecSys, the integration with A-LLMRec, which leverages the modality of item text information and the intensive capabilities of LLM, reduces this performance gap.

5.4.4 Beyond Recommendation: Language Generation Task (Favorite genre prediction). To validate whether A-LLMRec can generate natural language outputs based on the understanding of users and items through the aligned collaborative knowledge from CF-RecSys, we conduct a favorite genre prediction task (Figure 4). That is, given the same prompt format, we ask the LLM-based models (i.e., A-LLMRec and LLM-Only) using the same backbone LLM, which is OPT-6.7B, to predict the movie genres that a given user would enjoy watching. The only difference in the prompt is that while LLM-only is only given titles of movies watched by the user in the past, A-LLMRec is given the user representation and item embeddings along with the movie titles. In Figure 4, we observe that A-LLMRec indeed generates proper answers, while LLM-Only fails to do so. We attribute this to the fact that the item embeddings of the CF-RecSys are well aligned with the token space of the LLM, which enables the LLM to understand and utilize collaborative knowledge. Note that although we also experimented with TALLRec, we were not able to obtain valid outputs. We conjecture that since the LLM in TALLRec is fine-tuned via an instruction-tuning process that makes the model provide responses as part of the recommendation task, generating valid natural language outputs has become a non-trivial task. Please refer to Appendix B for the results of TALLRec.

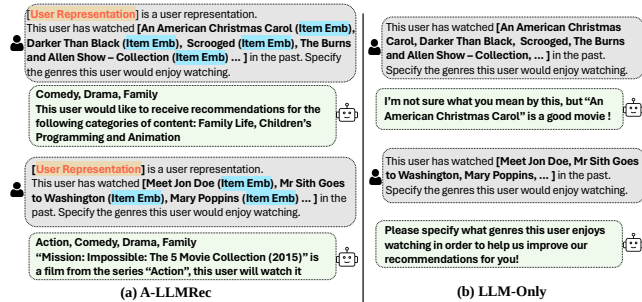
6 CONCLUSION

In this paper, we propose a novel LLM-based recommender system, named A-LLMRec. The main idea is to enable LLMs to utilize the collaborative knowledge from pre-trained CF-RecSys. By doing so, A-LLMRec outperforms existing CF-RecSys, modality-aware recommender systems, and LLM-based recommenders under various scenarios including cold/warm items, cold user, few-shot, and cross-domain scenarios. Moreover, we also demonstrate that the two advantages originated from fine-tuning neither pre-trained CF-RecSys nor LLMs, i.e., Model-agnostic and efficiency. Lastly, we show the potential of A-LLMRec in generating natural language tasks based on the understanding of collaborative knowledge from CF-RecSys. For future work, we plan to further enhance the ability of the LLM in A-LLMRec based on advanced prompt engineering such as chain-of-thought prompting [46].

Ethics Statement To the best of our knowledge, this paper aligns with the KDD Code of Ethics without any ethical concerns. The datasets and codes employed in our research are publicly available.

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**Figure 4: A-LLMRec v.s. LLM-Only on the favorite genre prediction task (Movies and TV dataset used).**